

Aarti Kumari

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Education

University of Massachusetts Amherst

Expected: December 2026

M.S. in Computer Science

GPA: 3.95

- Coursework: Machine Learning, Reinforcement Learning, Artificial Intelligence, Trustworthy & Responsible AI, Applied Statistics, Algorithms for Data Science, Database Implementation & Design, Computing for Digital Biomarkers in Healthcare.

West Bengal College of Engineering and Technology, India

August 2012 – May 2016

B.Tech in Electronics & Communications Engineering

GPA: 8.7/10.0

- Coursework: OOPS, DBMS, Java, C, Data Structures, Design Patterns, Satellite Communication

Technical Skills

Programming & Scripting: Java, Python, R, Shell Scripting (Linux/UNIX), OOP, Data Structures & Algorithms, Design Patterns, Ansible

ML/AI: Machine Learning, Temporal Graph Neural Networks, GraphSAGE, Deep Learning, Neural Networks, Reinforcement Learning, GenAI, NLP, RAG, SLM, LLM

Frameworks & Tools: Microservices; Search infrastructures, indexing pipelines; PyTorch, TensorFlow, Scikit-learn, FastAPI/Flask; Apache Spark, Elasticsearch, Lucene, Solr; REST API, GraphQL, Spring Boot, Docker, Jenkins, Git, Splunk, Datadog, Concurrency, Distributed Systems, CI/CD

Databases: MySQL, PostgreSQL, Hive, Snowflake, MongoDB; Query Optimization, Data Pipelines, Block Nested Loop, Buffer Manager

Industry Experience

Dell Technologies

Hopkinton, MA

Graduate AI Intern (Incoming)

Summer 2026

- Applying expertise in **AI, ML, LLMs** and **Distributed Systems** to drive next generation AI-powered product initiatives and research within Dell's AI & Data organization.

PayPal

Bengaluru, India

Software Engineer 2

September 2022 – August 2024

- Re-engineered post-payment pipelines using **multi-threaded concurrency**, reduced latency from **10X to 3X** with improved throughput and fault tolerance for **436M+** active accounts; boosted integration test coverage to **95%**.
- Led **NFC payment** backend for the wallet domain via **REST APIs & Spring Boot**; enabled secure **offline transactions** and reinforced payment security, driving **12%** revenue gains across Europe.
- Built **Top Contacts Recommendation** system using Learning-to-Rank (**LTR**) models, improving prediction accuracy by **86%**.
- Received **formal recognition** from PayPal Director for Send-Money core services optimization; enhanced **27%** system efficiency.
- Inducted, **mentored** and **trained 4 juniors** on Send-Money end to end flows to strengthen PayPal's live support team.

Altimetrik

Bengaluru, India

Senior Software Engineer

June 2021 – August 2022

- Implemented **six post-payment modules** with improved fault tolerance, throughput and performance by **60%**.
- Delivered Collections and Risk APIs and integrated ML models, including a **Delinquency Prediction Model** by using **Spring Boot** and **REST APIs** to flag high-risk accounts and enable early interventions with **96% accuracy**.

Dell EMC

Bengaluru, India

Software Developer

May 2017 – June 2021

- Engineered **Voyager** platform provisioning using Python, Ansible, and iDRAC APIs with a **92% reduction** in configuration time, enabling faster customer on-boarding, and earned the **2020 Spot Award** for delivering the Voyager fast-configuration backend.
- Validated SRM solution packs (MS-SQL, Oracle-SQL, MySQL, Hypervisor) via **Dockerized Jenkins CI/CD** pipelines cutting manual QA effort by **75%** and allowing continuous nightly builds.

Research & Projects

Research Assistant, UMass Amherst (Prof. Marco Serafini) 🐙 [GitHub](https://github.com)

June 2025 – Present

- Built Relational DB-to-Temporal Graph pipelines on Rel-Bench (Rel-Amazon **24M+** rows) with efficient sampling for time-stamped graph conversion and faster predictions.
- Developed dynamic graph updates, **memory-efficient GNN training**; benchmarked **GraphSAGE vs TGNNs** achieving **15-17% accuracy gain**.

AI Research Extern, IBM – 698DS UMass Industry Mentorship Program 🐙 [GitHub](https://github.com)

January 2026 – Present

- Researching **Open-Ended DPR Generation in the Wild** — designed multi-stage pipeline: clustering-based table discovery, LLM-driven DPR synthesis, SQL grounding with execution feedback, and metric-based evaluation (coverage, diversity, complexity, surprisal) over large heterogeneous data lakes.
- Strengthen SQL reliability via execution-aware self-correction, grounding validation, and hallucination mitigation across **500+ table** scale experiments; targeting combined metric score above **0.80**.

UAI Dream AI Hackathon – Finalist (Cambridge, June 2025) 🐙 [GitHub](https://github.com)

- Led design of **AI-powered matchmaking platform** (Spring Boot REST APIs, React, Flask Microservices) with **OCR (Tesseract)** for KYC, **CNN embeddings (90% accuracy)** for face validation, and **Transformer NLP + Jaccard similarity** for compatibility scoring.
- Reinforced security with GPS validation, Maps API, and Twilio real-time alerts; awarded **Top 10 Finalist** among 100+ teams.